Waste Right

The problem: household waste needs a simpler collection route

Scan. Confirm. Get it collected.

Mixed waste, human checks
and proposed local processing hubs



Pain points

Identification alone does not arrange collection

Residents

Mixed packaging is confusing.
Booking takes too many steps.
Final weight and fees are unclear.

Collection partners

Jobs need organising.
Materials and weight need checking.
Earnings and dues need clear records.

Our solution

One record connects the resident, helper and Operations

Scan and confirm	Phone camera or description. Correct or remove mistaken suggestions.
Book and collect	Choose a pickup. The helper verifies material and records actual weight.
Review and repeat	Accept the receipt, resolve issues and earn scan-day pickup credits.

The complete collection route

Household → collection → proposed hub → suitable destination

Receive and segregate	Accept segregated or mixed loads. Unsegregated service needs no home sorting.
Process suitable organics	Plan hub composting where feasible, or hand over to a processing partner.
Route other materials	Send suitable recyclables for recovery; isolate rejected and specialist waste.

Hub facilities and processing agreements are proposed; quality and capacity need validation.

Plans and measured charges

Monthly passes include segregation; Basic is pay per pickup

	Basic	Gold	Platinum
Price	₹10 / pickup	₹399 / month	₹469 / month
Pickups	1	14 / month	24 / month
Included per pickup	2 kg	8 kg	10 kg

Extra weight: ₹3/kg above the included allowance

Basic/reward mixed sorting: ₹5/kg. Evening fees: ₹5–₹7.
Helper weighs; resident reviews the measured receipt.

AI assistance, human verification

The app and trained research model have distinct roles

Main app

Image understanding + packaging OCR
Resident corrects the suggestions
Helper checks material and weight

Research model

450 labelled images
67 held-out test images
66.13% foreground mean IoU

IoU is region overlap, not item accuracy. One-household data; phone validation remains.

Sustainability and scale

Measure recovery, then repeat a working locality model

Quality and impact	Prompt hub sorting, controlled composting and records of accepted and rejected kg.
Local expansion	Dense routes, sufficient hub capacity, clear helper pay and measured operating costs.
Public programmes	Explore GOBARdhan processing partnerships and eligible NAMASTE support.

Proposed partnerships. No government award or quantified carbon saving is claimed.

Next: prove the locality economics

A four-week pilot with 50 households and two helpers

Prototype built

Scans, bookings, helper records, receipts and disputes

Revenue: pickup fees, subscriptions and actual recovery

₹10 fee − ₹8 base helper pay = ₹2 before other costs

Seeking a pilot location and processing arrangement